Python Application Programming

Assignment - 1

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Exercise 1.1

1. output:

   No way

2. output:

   No thanks, I can find something better.

3. output:

   I'll take it!

4. output:

   So long, suckers! I'll take it!

5. output:

   No thanks, I can find something better.

Exercise - 1.2

1. 10
   9
   8
   7
   6
   5
   4

2. 0.5
   0.5
   0.5
   0.5
   0.5

3. 10
   9
   8
   7

4. Letter # 0 is S
   Letter # 1 is n
   Letter # 2 is o
   Letter # 3 is w

Exercise - 1·3

1.

Iteration	n	i at print	what happens afterwards
1	10	10	10 % 2 = 5
2	10	5	5 + 1 = 6
3	10	6	6 % 2 = 3
4	10	3	3 + 1 = 4
5	10	4	4 % 2 = 2
6	10	2	2 % 2 = 1
7	10	1	1 + 1 = 2
8	10	2	2 % 2 = 1
9	10	1	1 + 1 = 2
10	10	2	2 % 2 = 1
...	10	1, 2	repeats forever

2. Ben appears to be trying to repeatedly reduce i : if i is even, divide it by 2 ; if it is odd, decrease it so that the loop can eventually reach zero.

The problem is that the code uses:

```
i = i + 1
```

when i is odd. when i becomes 1, this changes it to 2. Then 2 is divided by 2 to become 1 again, creating an infinite loop

one possible correction:

```
n = 10
i = 10
while i > 0:
    print (i)
    if i % 2 == 0:
        i = i // 2
    else:
        i = i - 1
```

Exercise - 1.4

```
amount = float ( input ( "Enter purchase amount ( in $): " ))
if amount >= 500 $ :
        discount = amount * 0.20
elif amount >= 200 $ :
        discount = amount * 0.10
else :
    discount = 0

final_price = amount - discount
print ( "Discount: " , discount
print ( "Final price: ", final_price )
print ( "Discount applied" if discount > 0 else "No discount" )
```

Test cases

Test case 1:

Input : 600 $

output

Discount: 120 $

Final price : 480 $

Discount applied

Test case 2:

Input: 300 $

output

Discount : 30 $

Final price : 270 $

Discount applied

Test case 3:

Input : 150 $

output

Discount : 0 $

Final price: 150 $

No discount

Exercise - 1.5

```
while True :
      password = input ( "Enter password: " )

      has_upper = any ( char. isupper () for char in password )
      has_lower = any ( char. islower () for char in password)
      has_digit = any ( char. isdigit () for char in password)

      if len ( password) >= 8 and has_upper and has_lower and
      has_digit :
          print ( "Valid" )
          break
      else:
          print ( "Invalid" )
```

Test cases

Test case 1

Input

abc123

output

Invalid

Test case 2

Input

SecurePass1

output

Valid

Test case 3

Input

passWORD2

output

Valid